

ArogyaRakshak — Dataset & Data-Source Research

Compiled for guide review · covers all 16 links shared in the WhatsApp group + additional sourcing per module (BillNyay / SchemeSetu / DawaCheck / Kadi)

0. Summary verdict (same conclusion Viraj reached, backed with sources)

Data need	Availability	Strategy
Government scheme data	Abundant, well-structured	Direct ingestion
Hospital bills / OCR source images	Limited but real datasets exist	Use directly + synthetic augmentation
Insurance claims (structured, non-Indian)	Abundant	Use for schema/ML training, adapt to Indian context
Insurance claims (structured, India-specific)	Thin	Use IRDAI/Dataful aggregate stats + the few Kaggle sets found
Policy brochures / wordings (unstructured PDFs)	Available raw, not parsed	Need PDF/table-extraction pipeline (own "jugaad")
Policy denial documents (actual letters)	Not available as real documents	Reverse-engineer: take structured denial-reason data (CARC/RARC codes, IRDAI grievance categories) → template-generate synthetic denial PDFs → use as test fixtures → feed back through BillNyay's own parser to validate the round-trip

1. Links you sent — analysed one by one

1.1 Hospital bills / patient records

#	Link	What it actually is	Usable for	Notes
1	Kaggle: Hospital Patient Data Records Dataset	984 anonymized tabular hospital patient encounters	BillNyay schema design, test fixtures	Structured, not OCR/image — good for DB modelling, not for OCR training
4	HuggingFace: naazimsh02/medocr-vision-dataset	2,462 image+text pairs , 59.4% medical (handwritten prescriptions, lab reports, OMR forms incl. Indian hospitals — Fortis, Manipal, Aakash Pathlab, Chethana Hospitals) + 40.6% general receipts/invoices (deliberately mixed in to prevent catastrophic forgetting)	Directly usable for OCR/vision-LLM fine-tuning in BillNyay's OCR+RAG pipeline	MIT license. 80/10/10 train/val/test split, 509 MB. Built by combining chinmays18/medical-prescription-dataset , saurabh1896/OMR-scanned-documents , dikshaasinghhh/bajaj (Kaggle lab reports) and mychen76/invoices-and-receipts_ocr_v1 . This is your single best find — go straight to the sub-sources too, since they're bigger individually

Additional hospital/patient datasets found (not in original list):

Dataset	Description	Link
chinmays18/medical-prescription-dataset	Source of the 1,000 handwritten prescriptions used above — larger on its own	huggingface.co/datasets/chinmays18/medical-prescription-dataset
dikshaasinghhh/bajaj (Kaggle)	Source of the 426 printed lab reports	kaggle.com/datasets/dikshaasinghhh/bajaj
mychen76/invoices-and-receipts_ocr_v1	2,043 invoice/receipt image-text pairs, general OCR pretraining	huggingface.co/datasets/mychen76/invoices-and-receipts_ocr_v1
SPARCS Hospital Inpatient Discharges (NY State, via Kaggle)	Real (US) de-identified discharge-level billing/diagnosis records — good structural reference for what a "real" claims table looks like	kaggle.com/datasets/bhautikmangukiya12/hospital-inpatient-discharges-dataset
KushagraWadhwa/medical-prescription-ocr-india (HF model, Apr 2026)	Fine-tuned specifically on Indian prescriptions — worth studying its dataset card for more India-specific OCR source data	huggingface.co/KushagraWadhwa/medical-prescription-ocr-india

1.2 Insurance claims (structured/tabular)

#	Link	What it is	Notes
2	<u>Kaggle: Indian Health Insurance Claims Analysis</u>	India-labelled claims dataset (page is JS-gated, couldn't verify row count directly — download and inspect)	Verify licensing before use
13	<u>Kaggle: Health Insurance Dataset India 2022</u>	Premium, scheme, location-level insurance data	Good for SchemeSetu eligibility-matching logic

Additional insurance-claims datasets found:

Dataset	Description	Link
Enhanced Health Insurance Claims Dataset	4,500 synthetic claims, ML-ready	kaggle.com/datasets/leandrenash/enhanced-health-insurance-claims-dataset
Insurance Claim Analysis: Demographic and Health (thedevastator)	Demographics → claim risk/severity	kaggle.com/datasets/thedevastator/insurance-claim-analysis-demographic-and-health
Healthcare Fraud Detection Dataset (nudratabbas)	10,000 claims with ICD-10 and CPT codes — closest thing to real billing-code structure available publicly	kaggle.com/datasets/nudratabbas/healthcare-fraud-detection-dataset
Sample Insurance Claim Prediction Dataset (easonlai)	Classic benchmark set	kaggle.com/datasets/easonlai/sample-insurance-claim-prediction-dataset
Indian Medical Insurance Policy (kethavathsamba)	India-specific policy-level data	kaggle.com/datasets/kethavathsamba/indian-medical-insurance-policy
Mendeley <u>insurance_claims</u> (fraud detection)	39-column claims-fraud dataset, well documented reproducibility steps	data.mendeley.com/datasets/992mh7dk9y
Dataful — Insurance category	Year & state-wise Claim Settlement under Health Insurance Business, Claims Development & Ageing (ex travel/PA) — this is real Indian aggregate regulatory data, not synthetic	dataful.in/datasets/?q=insurance+claims

1.3 Policy-denial data (your key blocker — solved via reverse engineering)

You were right that **no direct corpus of real denial letters/EOBs exists publicly** (PHI-protected everywhere). But **structured denial-reason taxonomies** do exist, and this is

exactly what you need to reverse-engineer synthetic denial documents:

Source	What it gives you	Link
X12 CARC/RARC codes (Claim Adjustment Reason Codes / Remittance Advice Remark Codes)	150+ standardized codes with reason text ("non-covered service," "exceeds benefit limit," "prior authorization required," etc.) — the international EOB/denial-reason standard	Overview: scribd.com/document/209319631/Reason-Codes (search "X12 CARC RARC full code list" for the canonical maintained list at x12.org)
IRDAI Bima Bharosa / Annual Report grievance tables	Real aggregate Indian data: FY25 had 2,57,790 grievances; 69% of general-insurance grievances were claims-related (delays/underpayment/rejection); category-wise breakdowns by insurer	irdai.gov.in (Annual Report PDF), joinditto.in/articles/general/irdai-annual-report
CGHS rate lists (see §1.4)	Gives you the "expected" reimbursable amount per procedure — combine with a claimed amount to synthetically generate a plausible "underpayment" or "excess charge" denial	see below

Your reverse-engineering pipeline (as you described, formalized):

1. Structured reason (CARC code / IRDAI category / CGHS rate mismatch) →
2. Template-fill into a realistic denial-letter/EOB **PDF** (using real insurer letterhead formats you can find via image search, redacted) →
3. Feed that PDF through BillNyay's own OCR+parsing pipeline →
4. Assert the round-trip recovers the original structured reason →
5. This becomes your **test/eval harness**, and doubles as a demo dataset for the guide.

1.4 Government schemes (your strongest category — confirmed)

#	Link	What it is	Notes
8	CGHS Rate List Login (tier III, NABH, private ward)	Dynamic/session-gated portal — can't be scraped directly	Needs manual export or automation with session handling; flag as a technical risk
9	CGHS Rate List Login (generic)	Same as above	—
10	CSIR mirror — CGHS Rates PDF	Static PDF mirror of CGHS package rates — this is the workaround for #8/#9	Use this — no login wall, directly parseable
11	myscheme.gov.in	National scheme search portal — 900+ Central + State schemes, eligibility criteria, benefits	Has a documented API used internally by the site (inspect network calls); large win for SchemeSetu
12	igod.gov.in — org list	India Government Open Data catalogue, org-wise dataset listing	Browse and download relevant health/insurance datasets directly
14	data.gov.in — Insurance dataset group	Official OGD platform insurance dataset group	Cross-check against Dataful for overlap

Additional scheme/government sources:

Source	Notes
PMJAY official (pmjay.gov.in)	Beneficiary verification API + hospital empanelment lists — distinct from MJPJAY per your earlier note
jeevandayee.gov.in (MJPJAY)	Marathi-native source, as already flagged in project memory
National Health Authority (NHA) open data	Ayushman Bharat implementation stats
NPPA / DPCO Schedule-I formulations	nppa.gov.in — ceiling prices for ~928 scheduled formulations, revised annually on WPI; confirms your existing technical note that this must be pulled live, not hardcoded

1.5 Policy documents / brochures (unstructured)

#	Link	What it is	Notes
16	Testbook — Insurance Companies PDF	List of Indian insurance companies (likely exam-prep content)	Low-value as data, but useful as a company/insurer master-list seed

For actual **policy wordings/brochures**, no single link was sent — insurers (Star Health, HDFC Ergo, ICICI Lombard, Niva Bupa, etc.) publish these as PDFs on their own sites. This is the "jugaad" category you flagged: needs a **PDF table + clause extraction pipeline** (your `pdf` skill / layout-aware parsing), not a dataset download. Treat as an ingestion-pipeline problem, not a data-sourcing problem.

1.6 Data marketplaces / aggregators (browse, don't ingest directly)

#	Link	Verdict
6	Datarade — Hospital data category	Commercial data marketplace — mostly paid vendor listings (Dun & Bradstreet-style hospital directories, US-centric). Useful only to see what schema/fields "real" commercial hospital data providers structure around — not a free source
3	HDX — DHS Subnational Data for India	Demographic and Health Survey (DHS) subnational indicators — real, free, granular Indian health indicators (state/district level)
3b	HDX — DHS resource: 60b5095a...	Specific DHS resource file — same as above, download directly
5	Dataful — medical expenses search	Aggregator with curated Indian govt datasets on health spending, out-of-pocket expenditure, insurance claims (already cited in §1.2)

1.7 Research literature

#	Link	Status
7	PMC9664599	Blocked by reCAPTCHA on fetch — re-check manually , exact title unconfirmed in this pass. Related literature found instead (below) is directly relevant and should be cited in your Phase-0/1 documentation regardless

Related papers worth citing in your report (found during this search):

- *India's Largest Hospital Insurance Program Faces Challenges In Using Claims Data To Measure Quality* — Morton, Nagpal, Sadanandan, Bauhoff, *Health Affairs* 2016

(PMC7473072) — directly relevant to BillNyay's core premise

- *Awareness of India's national health insurance scheme (PM-JAY): a cross-sectional study across six states* — Parisi et al., *Health Policy Plan* 2022 (PMC10019566) — relevant to SchemeSetu's low-awareness problem statement
- *Public Health Insurance Status and Utilization of Healthcare Services Across India: A Narrative Review* — Shende & Wagh, *Cureus* 2024 (PMC10944651)
- *An Audit of Indian Health Insurance Claims for Mental Illness from Pooled IIB Macroindicator Data* — Mohandoss & Thavarajah, *Indian J Psychol Med* 2017 (PMC5461833) — good example of what's achievable with **IIB (Insurance Information Bureau of India)** pooled data
- **Insurance Information Bureau of India (IIB)** itself (iib.gov.in) — worth investigating directly; it's the body that aggregates claims macrodata across all Indian insurers and is the source several of the above papers used

2. New datasets found beyond your list (by module)

BillNyay (bill auditing / appeal generation)

Dataset	Why it matters
National Health Claims Exchange (NHCX) — NHA's 2024 unified claims portal	Understand the <i>target</i> data format (FHIR-based) your outputs should eventually be compatible with
X12 837/835 claim & remittance format documentation	The actual EDI schema real claims/denials use in most digitized systems — useful schema reference even though India uses NHCX/FHIR now
IIB (Insurance Information Bureau of India) pooled claims macrodata	Closest thing to real aggregate Indian claims data; may require registration/access request

SchemeSetu (eligibility)

Dataset	Why it matters
myscheme.gov.in API (network-inspectable)	Full scheme + eligibility criteria corpus
PMJAY hospital empanelment + beneficiary datasets	Core to national scheme matching
DHS India subnational indicators (HDX)	Demographic eligibility heuristics

DawaCheck (medicine pricing)

Dataset	Why it matters
NPPA ceiling price list (Schedule-I formulations, live)	Already a confirmed technical note — reconfirmed still accurate as of the 143rd Authority meeting (Feb 2026)
Jan Aushadhi (PMBJP) price list	Generic-medicine price comparator — not in your original list, high-value addition for DawaCheck's "transparency" angle

Kadi (entity resolution)

Dataset	Why it matters
Hospital/insurer name master lists (from Testbook PDF + IIB + IRDAI registered-insurer list)	Ground truth entity list to test transliteration/fuzzy-matching against

3. Consolidated action list for tomorrow's meeting

1. Show the guide this table as evidence of the "available vs. reverse-engineered" split you already reasoned through on WhatsApp.
2. **Prioritize downloading:** [naazimsnh02/medocr-vision-dataset](https://naazimsnh02.github.io/medocr-vision-dataset/) (and its 4 sub-sources) tonight — it's the biggest concrete win and directly OCR-ready.
3. Flag CGHS rate-list login pages (#8, #9) as a technical blocker; use the CSIR static PDF mirror (#10) as the workaround.
4. Formally document the denial-data reverse-engineering pipeline (CARC/RARC codes → synthetic PDF → OCR round-trip) as your answer to teammate concern # (OCR accuracy / authenticity) from the 16-point list already in project docs.
5. Investigate IIB (iib.gov.in) access — it's the one real Indian claims-macrodata source multiple cited papers used, and nobody in the group has looked at it yet.